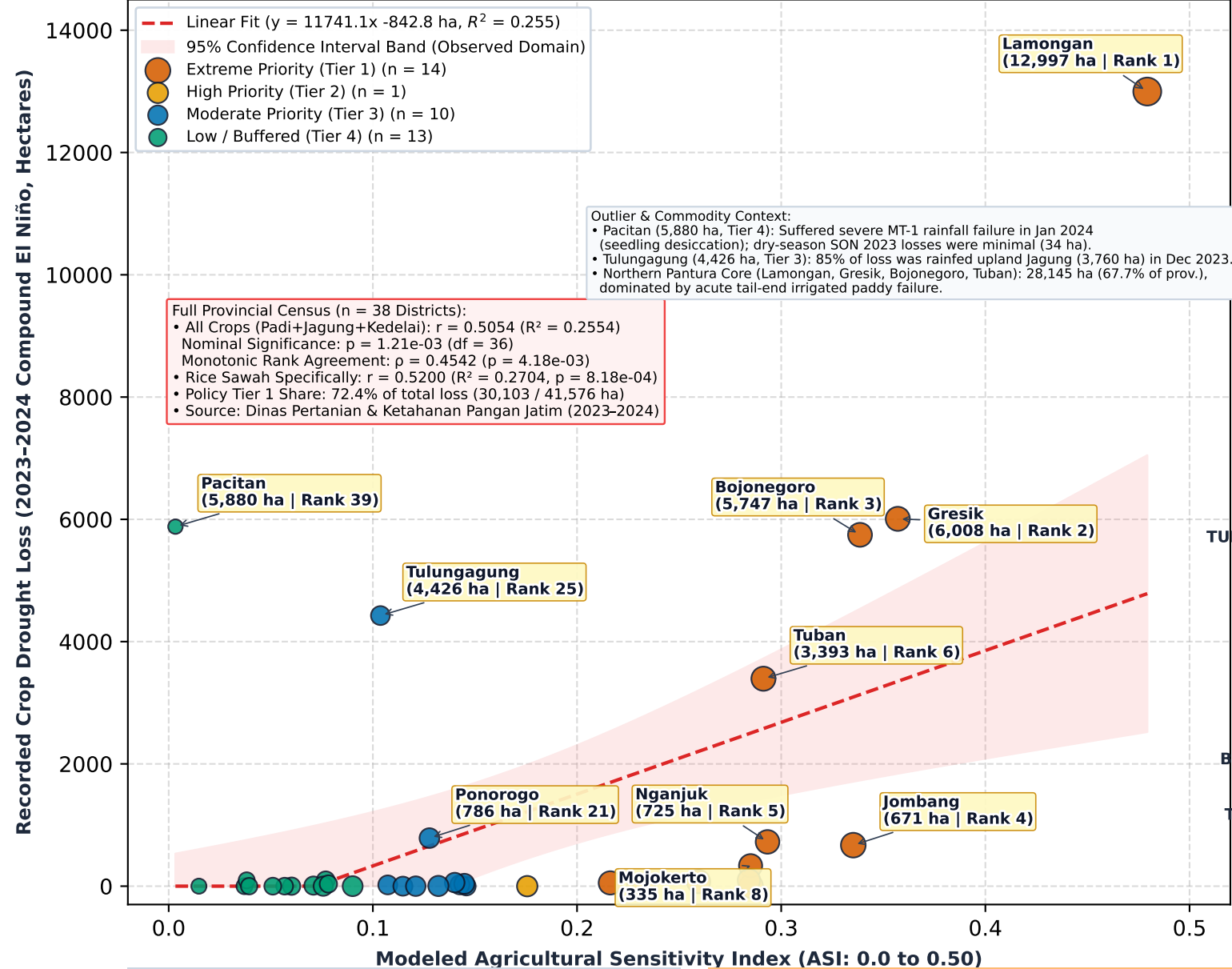


Empirical Agricultural Disaster Ground-Truth Validation (2023-2024 Compound El Niño)

Macro-Validation of Modeled Agricultural Sensitivity Index (ASI) vs Official Dinas Pertanian & Ketahanan Pangan Jawa Timur Crop Loss Records | Full Provincial Census (n = 38 Districts, 24-Month Compound Cycle)

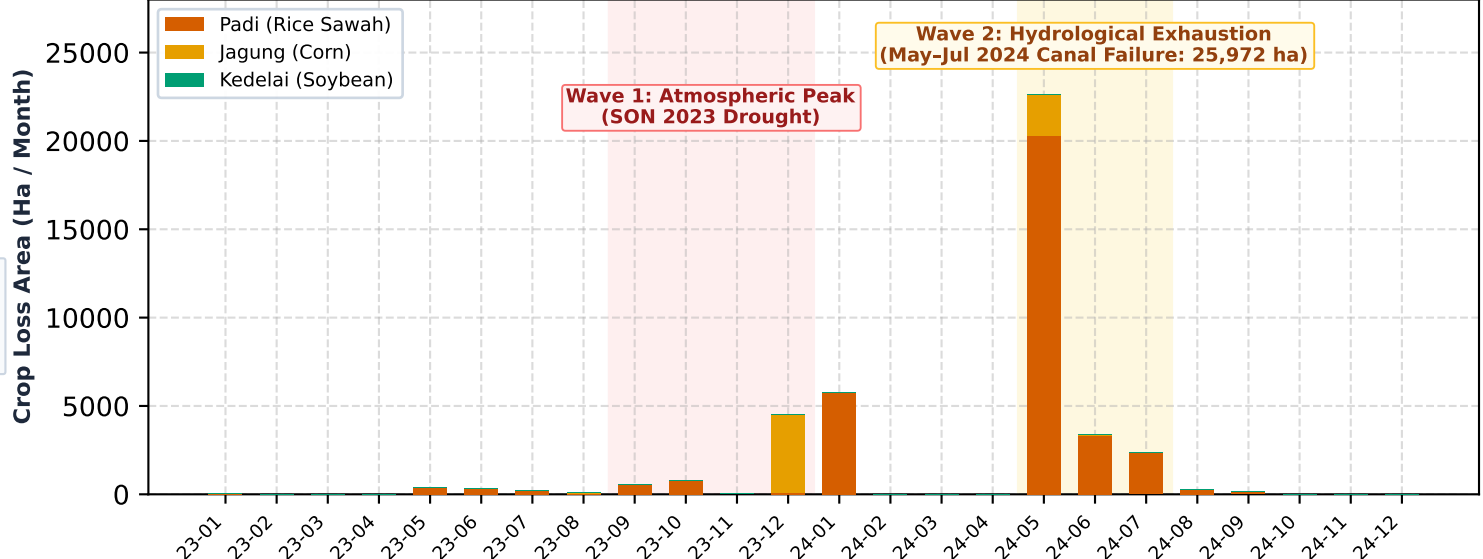
(a) Macro Statistical Concordance & Exposure Correlation

Modeled Agricultural Sensitivity Index (ASI) vs Recorded Crop Drought Loss (2023-2024)



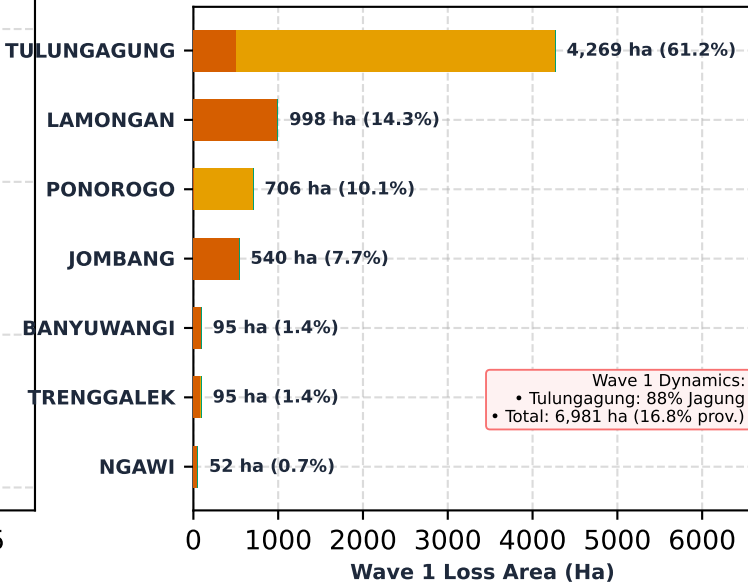
(b) 24-Month Temporal Evolution: Dual-Wave Compound Disaster Progression (Jan 2023 - Dec 2024)

Primary Meteorological Deficit (2023) Followed by Delayed Monsoon & Reservoir Depletion Spike (May 2024)



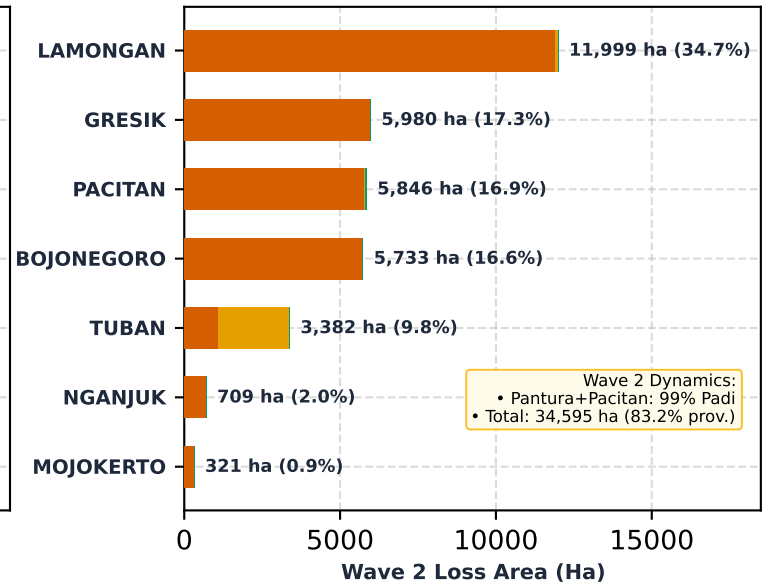
(c) Wave 1 (2023) Epicenters (6,981 ha)

Atmospheric Deficit & Upland Corn Drought



(d) Wave 2 (2024) Epicenters (34,595 ha)

Canal Drawdown & MT-1 Rain Failure



Full Census Ground Truth

Evaluated across all 38 regencies/cities in East Java (Dinas Pertanian Jatim official data), modeled ASI significantly accounts for crop drought losses ($r = 0.5054$, $p = 0.0012$, $\rho = 0.4542$). Rice sawah specifically demonstrates even higher concordance ($r = 0.5200$, $p = 0.0008$).

Policy Tier 1 Concentration

Districts categorized as Extreme Priority (Tier 1) in Milestone 6 encompass 72.4% (30,103 ha) of all recorded agricultural drought losses in East Java, validating that our policy intervention matrix targets the true operational disaster epicenters.

Dual-Wave Disaster Progression

The 2023-2024 compound El Niño displayed two distinct phases: an initial meteorological deficit during SON 2023, followed by a massive secondary canal drawdown spike in May 2024 (22,668 ha in 1 month), driven by exhausted reservoir storage and delayed monsoon recharge.

Tail-End Canal Vulnerability

Over 67.7% of all provincial crop damage concentrated in the lower Bengawan Solo and karst corridor (Lamongan, Gresik, Bojonegoro, Tuban). Irrigated rice sawah accounted for 83.1% of all damage, highlighting tail-end canal rationing as the principal operational vulnerability.